

NETWORK FOUNDATIONS

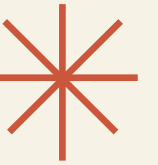


THE INTERNET AND NETWORK FUNDAMENTALS



WHAT *IS* A COMPUTER NETWORK?

COLLECTION OF DEVICES, ROUTERS, AND LINKS, ALL OF WHICH IS MANAGED BY AN ORGANIZATION OF SORTS, FOR THE SAKE OF RUNNING A COMPUTER.



Why is the internet a “network of networks”?

Made up of Interconnected ISPs, each of which is a network

How do we interact with computer networks?

Making up the Network’s edge

How do billions of devices communicate through interconnected networks?

Various protocols sending info across the internet

What are End Systems/Hosts?

PC, Smart Phone, Smart Watch, etc.

NETWORK EDGE AND NETWORK CORE



NETWORK EDGE VS NETWORK CORE

How one accesses
the internet

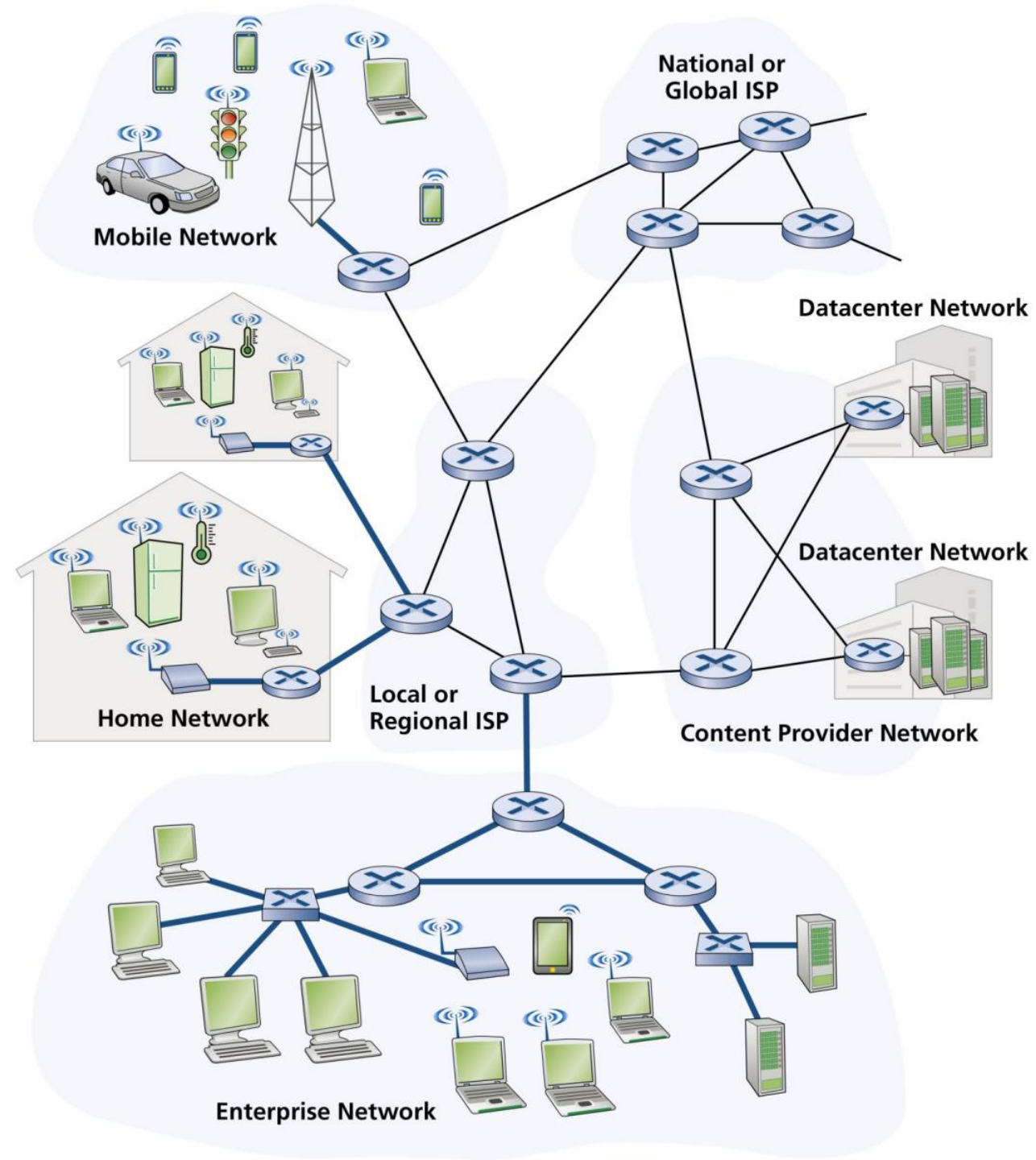
VS

The internet and it's
protocols



NETWORK EDGE

- Hosts
- Clients
- Servers
- Access Networks
 - Home Networks
 - Enterprise Networks
 - Mobile Networks





NETWORK CORE

- Routers
- Switches
- Links
- Packet Forwarding
- Packet Switching
- Circuit Switching

NETWORK DIAGRAM

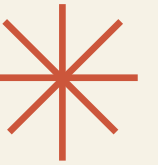


COMMUNICATION AND NETWORK PROTOCOLS



WHAT IS A NETWORK PROTOCOL?

AND WHY ARE THEY NECESSARY?



HTTP/HTTPS –Transfer Protocol (Secure)

- Web Document Request
- Application Layer

DNS – Domain Name System

- Turns Host names into IP addresses
- Application Layer

TCP – Transmission Control Protocol

- Transports data, breaks long messages down, congestion control
- Transport Layer

IP – Internet Protocol

- Specifies the format of Packets
- Network Layer



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IN OUR
LIVES

UNDERSTANDING NETWORK PERFORMANCE



WHAT AFFECTS NETWORK PERFORMANCE?

- Throughput
 - The amount of data per second that can be transferred
- Bandwidth
 - The width of the frequency band
- Delay
 - The temporary halting of a packet's journey at various nodes.
- Transmission Delay
 - A Delay occurring to packets while waiting to their turn to transmit
- Queuing Delay
 - Delays to the arrival of the packets on account of network congestion
- Packet Loss
 - A packet is dropped due to a full buffer





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NETWORK SECURITY



WHAT DOES THIS INVOLVE

- Authentication
 - Making sure only specific sources can participate in a certain protocol
- Encryption
 - Scrambling a message so only those with it's key can read it
- Confidentiality
 - Preventing information from being seen by those without permission
- Network Threats
 - Threats to the network
- IP Spoofing
 - Injecting packets with fake source address
- Packet Sniffing
 - A passive receiver records a copy of passing packets



HOW TO PROTECT NETWORKS

- Firewalls
- VPN (Virtual Private Networks)
- Secure Authentication



THAT'S ALL
